

Detection and quantification of methicillin-resistant Staphylococcus aureus (MRSA) clones in retail meat products.

AIMS: The objective of the study was to determine the prevalence of methicillin-resistant Staphylococcus aureus (MRSA) contamination of retail meat and to determine the level of contamination. METHODS AND RESULTS: Pork (pork chops and ground pork), ground beef and chicken (legs, wings and thighs) were purchased at retail outlets in four Canadian provinces and tested for the presence of methicillin-resistant Staph. aureus using qualitative and quantitative methods. MRSA was isolated from 9.6% of pork, 5.6% of beef and 1.2% of chicken samples ($P = 0.0002$). Low levels of MRSA were typically present, with 37% below the detection threshold for quantification and <100 CFU g⁻¹ present in most quantifiable samples. All isolates were classified as Canadian epidemic MRSA-2 (CMRSA-2) by pulsed field gel electrophoresis (PFGE), with two different PFGE subtypes, and were spa type 24/t242. CONCLUSIONS: MRSA contamination of retail meat is not uncommon. While CMRSA-2, a human epidemic clone, has been found in pigs in Canada, the lack of isolation of livestock-associated ST398 was surprising. SIGNIFICANCE AND IMPACT OF THE STUDY: The relevance of MRSA contamination of meat is unclear but investigation is required because of the potential for exposure from food handling. Sources of contamination require investigation because these results suggest that human or animal sources could be involved.